

Question Bank

Topic: Domain and range

[Thomas Calculus]

1. $f(x) = 1 + x^2$

2. $f(x) = 1 - \sqrt{x}$

3. $F(x) = \sqrt{5x + 10}$

4. $g(x) = \sqrt{x^2 - 3x}$

5. $f(t) = \frac{4}{3 - t}$

6. $G(t) = \frac{2}{t^2 - 16}$

15. $f(x) = 5 - 2x$

16. $f(x) = 1 - 2x - x^2$

17. $g(x) = \sqrt{|x|}$

18. $g(x) = \sqrt{-x}$

19. $F(t) = t/|t|$

20. $G(t) = 1/|t|$

21. Find the domain of $y = \frac{x + 3}{4 - \sqrt{x^2 - 9}}$.

In Exercises 19–28, find the (a) domain and (b) range.

19. $y = |x| - 2$

20. $y = -2 + \sqrt{1 - x}$

21. $y = \sqrt{16 - x^2}$

22. $y = 3^{2-x} + 1$

23. $y = 2e^{-x} - 3$

24. $y = \tan(2x - \pi)$

25. $y = 2 \sin(3x + \pi) - 1$

26. $y = x^{2/5}$

27. $y = \ln(x - 3) + 1$

28. $y = -1 + \sqrt[3]{2 - x}$

Piecewise-Defined Functions

In Exercises 31 and 32, find the (a) domain and (b) range.

$$31. y = \begin{cases} \sqrt{-x}, & -4 \leq x \leq 0 \\ \sqrt{x}, & 0 < x \leq 4 \end{cases}$$

$$32. y = \begin{cases} -x - 2, & -2 \leq x \leq -1 \\ x, & -1 < x \leq 1 \\ -x + 2, & 1 < x \leq 2 \end{cases}$$

[Howard Anton]

$$(a) f(x) = \frac{x^2 + x}{x + 1}; g(x) = x$$

$$(b) f(x) = \frac{x\sqrt{x} + \sqrt{x}}{x + 1}; g(x) = \sqrt{x}$$

$$9. (a) f(x) = \frac{1}{x - 3}$$

$$(b) F(x) = \frac{x}{|x|}$$

$$(c) g(x) = \sqrt{x^2 - 3}$$

$$(d) G(x) = \sqrt{x^2 - 2x + 5}$$

$$(e) h(x) = \frac{1}{1 - \sin x}$$

$$(f) H(x) = \sqrt{\frac{x^2 - 4}{x - 2}}$$

$$10. (a) f(x) = \sqrt{3 - x}$$

$$(b) F(x) = \sqrt{4 - x^2}$$

$$(c) g(x) = 3 + \sqrt{x}$$

$$(d) G(x) = x^3 + 2$$

$$(e) h(x) = 3 \sin x$$

$$(f) H(x) = (\sin \sqrt{x})^{-2}$$

[Stewart]

$$23. f(x) = \frac{x}{3x - 1}$$

$$24. f(x) = \frac{5x + 4}{x^2 + 3x + 2}$$

$$25. f(t) = \sqrt{t} + \sqrt[3]{t}$$

$$26. g(u) = \sqrt{u} + \sqrt{4 - u}$$

$$27. h(x) = \frac{1}{\sqrt[4]{x^2 - 5x}}$$

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28. Find the domain and range and sketch the graph of the function $h(x) = \sqrt{4 - x^2}$.

29–40 |||| Find the domain and sketch the graph of the function.

$$29. f(x) = 5$$

$$30. F(x) = \frac{1}{2}(x + 3)$$

$$31. f(t) = t^2 - 6t$$

$$32. H(t) = \frac{4 - t^2}{2 - t}$$

$$33. g(x) = \sqrt{x - 5}$$

$$34. F(x) = |2x + 1|$$

$$35. G(x) = \frac{3x + |x|}{x}$$

$$36. g(x) = \frac{|x|}{x^2}$$

$$37. f(x) = \begin{cases} x & \text{if } x \leq 0 \\ x + 1 & \text{if } x > 0 \end{cases}$$

$$38. f(x) = \begin{cases} 2x + 3 & \text{if } x < -1 \\ 3 - x & \text{if } x \geq -1 \end{cases}$$

$$39. f(x) = \begin{cases} x + 2 & \text{if } x \leq -1 \\ x^2 & \text{if } x > -1 \end{cases}$$

[Graph sketching]

$$25. f(x) = \begin{cases} x, & 0 \leq x \leq 1 \\ 2 - x, & 1 < x \leq 2 \end{cases}$$

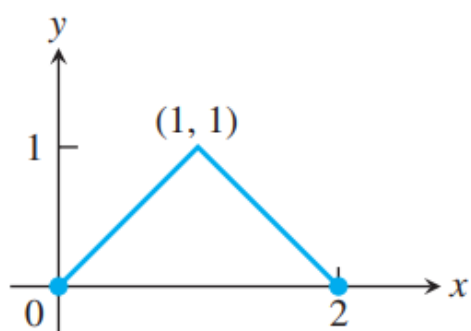
$$26. g(x) = \begin{cases} 1 - x, & 0 \leq x \leq 1 \\ 2 - x, & 1 < x \leq 2 \end{cases}$$

$$27. F(x) = \begin{cases} 4 - x^2, & x \leq 1 \\ x^2 + 2x, & x > 1 \end{cases}$$

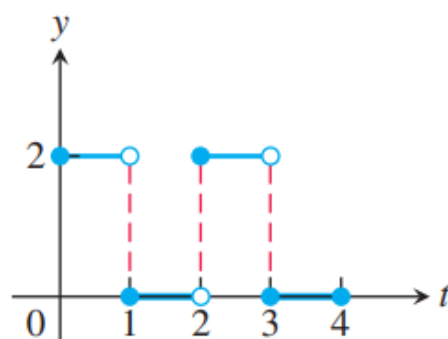
$$28. G(x) = \begin{cases} 1/x, & x < 0 \\ x, & 0 \leq x \end{cases}$$

Find a formula for each function graphed in Exercises 29–32.

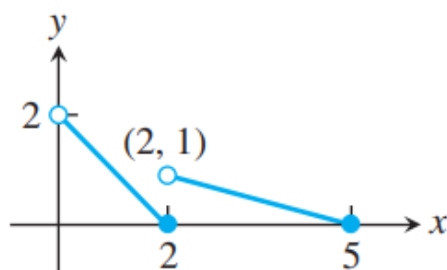
29. a.



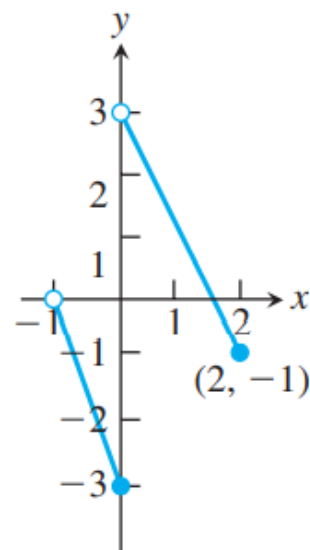
b.



30. a.



b.



Graph the functions in Exercises 35–54.

35. $y = \sqrt{x+4}$

37. $y = |x-2|$

39. $y = 1 + \sqrt{x-1}$

41. $y = (x+1)^{2/3}$

43. $y = 1 - x^{2/3}$

45. $y = \sqrt[3]{x-1} - 1$

47. $y = \frac{1}{x-2}$

49. $y = \frac{1}{x} + 2$

51. $y = \frac{1}{(x-1)^2}$

53. $y = \frac{1}{x^2} + 1$

36. $y = \sqrt{9-x}$

38. $y = |1-x| - 1$

40. $y = 1 - \sqrt{x}$

42. $y = (x-8)^{2/3}$

44. $y + 4 = x^{2/3}$

46. $y = (x+2)^{3/2} + 1$

48. $y = \frac{1}{x} - 2$

50. $y = \frac{1}{x+2}$

52. $y = \frac{1}{x^2} - 1$

54. $y = \frac{1}{(x+1)^2}$

[Howard Anton]

9. (a) $f(x) = \frac{1}{x-3}$

(c) $g(x) = \sqrt{x^2-3}$

(e) $h(x) = \frac{1}{1-\sin x}$

10. (a) $f(x) = \sqrt{3-x}$

(c) $g(x) = 3 + \sqrt{x}$

(e) $h(x) = 3 \sin x$

(b) $F(x) = \frac{x}{|x|}$

(d) $G(x) = \sqrt{x^2-2x+5}$

(f) $H(x) = \sqrt{\frac{x^2-4}{x-2}}$

(b) $F(x) = \sqrt{4-x^2}$

(d) $G(x) = x^3 + 2$

(f) $H(x) = (\sin \sqrt{x})^{-2}$

[Stewart]

$$23. f(x) = \frac{x}{3x - 1}$$

$$24. f(x) = \frac{5x + 4}{x^2 + 3x + 2}$$

$$25. f(t) = \sqrt{t} + \sqrt[3]{t}$$

$$26. g(u) = \sqrt{u} + \sqrt{4 - u}$$

$$27. h(x) = \frac{1}{\sqrt[4]{x^2 - 5x}}$$

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28. Find the domain and range and sketch the graph of the function $h(x) = \sqrt{4 - x^2}$.

29–40 |||| Find the domain and sketch the graph of the function.

$$29. f(x) = 5$$

$$30. F(x) = \frac{1}{2}(x + 3)$$

$$31. f(t) = t^2 - 6t$$

$$32. H(t) = \frac{4 - t^2}{2 - t}$$

$$33. g(x) = \sqrt{x - 5}$$

$$34. F(x) = |2x + 1|$$

$$35. G(x) = \frac{3x + |x|}{x}$$

$$36. g(x) = \frac{|x|}{x^2}$$

$$37. f(x) = \begin{cases} x & \text{if } x \leq 0 \\ x + 1 & \text{if } x > 0 \end{cases}$$

$$38. f(x) = \begin{cases} 2x + 3 & \text{if } x < -1 \\ 3 - x & \text{if } x \geq -1 \end{cases}$$

$$39. f(x) = \begin{cases} x + 2 & \text{if } x \leq -1 \\ x^2 & \text{if } x > -1 \end{cases}$$

New Functions from old

Algebraic Combinations

In Exercises 1 and 2, find the domains and ranges of f , g , $f + g$, and $f \cdot g$.

1. $f(x) = x$, $g(x) = \sqrt{x - 1}$
2. $f(x) = \sqrt{x + 1}$, $g(x) = \sqrt{x - 1}$

In Exercises 3 and 4, find the domains and ranges of f , g , f/g , and g/f .

3. $f(x) = 2$, $g(x) = x^2 + 1$
4. $f(x) = 1$, $g(x) = 1 + \sqrt{x}$

Composites of Functions

5. If $f(x) = x + 5$ and $g(x) = x^2 - 3$, find the following.

- a. $f(g(0))$
- b. $g(f(0))$
- c. $f(g(x))$
- d. $g(f(x))$

e. $f(f(-5))$

f. $g(g(2))$

g. $f(f(x))$

h. $g(g(x))$

6. If $f(x) = x - 1$ and $g(x) = 1/(x + 1)$, find the following.

a. $f(g(1/2))$

b. $g(f(1/2))$

c. $f(g(x))$

d. $g(f(x))$

e. $f(f(2))$

f. $g(g(2))$

g. $f(f(x))$

h. $g(g(x))$

In Exercises 7–10, write a formula for $f \circ g \circ h$.

7. $f(x) = x + 1$, $g(x) = 3x$, $h(x) = 4 - x$

8. $f(x) = 3x + 4$, $g(x) = 2x - 1$, $h(x) = x^2$

9. $f(x) = \sqrt{x + 1}$, $g(x) = \frac{1}{x + 4}$, $h(x) = \frac{1}{x}$

10. $f(x) = \frac{x + 2}{3 - x}$, $g(x) = \frac{x^2}{x^2 + 1}$, $h(x) = \sqrt{2 - x}$

Let $f(x) = x - 3$, $g(x) = \sqrt{x}$, $h(x) = x^3$, and $j(x) = 2x$. Express each of the functions in Exercises 11 and 12 as a composite involving one or more of f , g , h , and j .

11. a. $y = \sqrt{x} - 3$

b. $y = 2\sqrt{x}$

c. $y = x^{1/4}$

d. $y = 4x$

e. $y = \sqrt{(x - 3)^3}$

f. $y = (2x - 6)^3$

12. a. $y = 2x - 3$

b. $y = x^{3/2}$

c. $y = x^9$

d. $y = x - 6$

e. $y = 2\sqrt{x - 3}$

f. $y = \sqrt{x^3 - 3}$

Vertical and Horizontal Scaling

Exercises 57–66 tell by what factor and direction the graphs of the given functions are to be stretched or compressed. Give an equation for the stretched or compressed graph.

- 57. $y = x^2 - 1$, stretched vertically by a factor of 3
- 58. $y = x^2 - 1$, compressed horizontally by a factor of 2
- 59. $y = 1 + \frac{1}{x^2}$, compressed vertically by a factor of 2
- 60. $y = 1 + \frac{1}{x^2}$, stretched horizontally by a factor of 3
- 61. $y = \sqrt{x + 1}$, compressed horizontally by a factor of 4
- 62. $y = \sqrt{x + 1}$, stretched vertically by a factor of 3
- 63. $y = \sqrt{4 - x^2}$, stretched horizontally by a factor of 2
- 64. $y = \sqrt{4 - x^2}$, compressed vertically by a factor of 3
- 65. $y = 1 - x^3$, compressed horizontally by a factor of 3
- 66. $y = 1 - x^3$, stretched horizontally by a factor of 2

Exercises 25–34 tell how many units and in what directions the graphs of the given equations are to be shifted. Give an equation for the shifted graph. Then sketch the original and shifted graphs together, labeling each graph with its equation.

- 25. $x^2 + y^2 = 49$ Down 3, left 2
- 26. $x^2 + y^2 = 25$ Up 3, left 4
- 27. $y = x^3$ Left 1, down 1
- 28. $y = x^{2/3}$ Right 1, down 1
- 29. $y = \sqrt{x}$ Left 0.81
- 30. $y = -\sqrt{x}$ Right 3
- 31. $y = 2x - 7$ Up 7
- 32. $y = \frac{1}{2}(x + 1) + 5$ Down 5, right 1
- 33. $y = 1/x$ Up 1, right 1
- 34. $y = 1/x^2$ Left 2, down 1

Graphing

In Exercises 67–74, graph each function, not by plotting points, but by starting with the graph of one of the standard functions presented in Figures 1.14–1.17 and applying an appropriate transformation.

67. $y = -\sqrt{2x + 1}$

68. $y = \sqrt{1 - \frac{x}{2}}$

69. $y = (x - 1)^3 + 2$

70. $y = (1 - x)^3 + 2$

71. $y = \frac{1}{2x} - 1$

72. $y = \frac{2}{x^2} + 1$

73. $y = -\sqrt[3]{x}$

74. $y = (-2x)^{2/3}$

75. Graph the function $y = |x^2 - 1|$.

76. Graph the function $y = \sqrt{|x|}$.

In Exercises 37 and 38, **(a)** write formulas for $f \circ g$ and $g \circ f$ and find the **(b)** domain and **(c)** range of each.

37. $f(x) = 2 - x^2$, $g(x) = \sqrt{x + 2}$

38. $f(x) = \sqrt{x}$, $g(x) = \sqrt{1 - x}$

For Exercises 39 and 40, sketch the graphs of f and $f \circ f$.

39.
$$f(x) = \begin{cases} -x - 2, & -4 \leq x \leq -1 \\ -1, & -1 < x \leq 1 \\ x - 2, & 1 < x \leq 2 \end{cases}$$

40.
$$f(x) = \begin{cases} x + 1, & -2 \leq x < 0 \\ x - 1, & 0 \leq x \leq 2 \end{cases}$$

49. Suppose the graph of g is given. Write equations for the graphs that are obtained from the graph of g by shifting, scaling, or reflecting, as indicated.

a. Up $\frac{1}{2}$ unit, right 3

b. Down 2 units, left $\frac{2}{3}$

c. Reflect about the y -axis

d. Reflect about the x -axis

e. Stretch vertically by a factor of 5

f. Compress horizontally by a factor of 5

[Howard Anton]

5. $y = -2(x + 1)^2 - 3$

6. $y = \frac{1}{2}(x - 3)^2 + 2$

7. $y = x^2 + 6x$

8. $y = \frac{1}{2}(x^2 - 2x + 3)$

9. $y = 3 - \sqrt{x + 1}$

10. $y = 1 + \sqrt{x - 4}$

11. $y = \frac{1}{2}\sqrt{x} + 1$

12. $y = -\sqrt{3x}$

13. $y = \frac{1}{x - 3}$

14. $y = \frac{1}{1 - x}$

15. $y = 2 - \frac{1}{x + 1}$

16. $y = \frac{x - 1}{x}$

17. $y = |x + 2| - 2$

18. $y = 1 - |x - 3|$

19. $y = |2x - 1| + 1$

20. $y = \sqrt{x^2 - 4x + 4}$

21. $y = 1 - 2\sqrt[3]{x}$

22. $y = \sqrt[3]{x - 2} - 3$

23. $y = 2 + \sqrt[3]{x + 1}$

24. $y + \sqrt[3]{x - 2} = 0$

25. (a) Sketch the graph of $y = x + |x|$ by adding the corresponding y -coordinates on the graphs of $y = x$ and $y = |x|$.

(b) Express the equation $y = x + |x|$ in piecewise form with no absolute values, and confirm that the graph you obtained in part (a) is consistent with this equation.

37–42 Express f as a composition of two functions; that is, find g and h such that $f = g \circ h$. [Note: Each exercise has more than one solution.] ■

37. (a) $f(x) = \sqrt{x+2}$

(b) $f(x) = |x^2 - 3x + 5|$

38. (a) $f(x) = x^2 + 1$

(b) $f(x) = \frac{1}{x-3}$

39. (a) $f(x) = \sin^2 x$

(b) $f(x) = \frac{3}{5 + \cos x}$

40. (a) $f(x) = 3 \sin(x^2)$

(b) $f(x) = 3 \sin^2 x + 4 \sin x$

41. (a) $f(x) = (1 + \sin(x^2))^3$

(b) $f(x) = \sqrt{1 - \sqrt[3]{x}}$

42. (a) $f(x) = \frac{1}{1-x^2}$

(b) $f(x) = |5 + 2x|$

27–28 Find formulas for $f + g$, $f - g$, fg , and f/g , and state the domains of the functions. ■

27. $f(x) = 2\sqrt{x-1}$, $g(x) = \sqrt{x-1}$

28. $f(x) = \frac{x}{1+x^2}$, $g(x) = \frac{1}{x}$

29. Let $f(x) = \sqrt{x}$ and $g(x) = x^3 + 1$. Find

(a) $f(g(2))$ (b) $g(f(4))$ (c) $f(f(16))$

(d) $g(g(0))$ (e) $f(2+h)$ (f) $g(3+h)$.

30. Let $g(x) = \sqrt{x}$. Find

(a) $g(5s+2)$ (b) $g(\sqrt{x}+2)$ (c) $3g(5x)$

(d) $\frac{1}{g(x)}$ (e) $g(g(x))$ (f) $(g(x))^2 - g(x^2)$

(g) $g(1/\sqrt{x})$ (h) $g((x-1)^2)$ (i) $g(x+h)$.

31–34 Find formulas for $f \circ g$ and $g \circ f$, and state the domains of the compositions. ■

31. $f(x) = x^2$, $g(x) = \sqrt{1-x}$

32. $f(x) = \sqrt{x-3}$, $g(x) = \sqrt{x^2+3}$

33. $f(x) = \frac{1+x}{1-x}$, $g(x) = \frac{x}{1-x}$

34. $f(x) = \frac{x}{1+x^2}$, $g(x) = \frac{1}{x}$

35–36 Find a formula for $f \circ g \circ h$. ■

35. $f(x) = x^2 + 1$, $g(x) = \frac{1}{x}$, $h(x) = x^3$

36. $f(x) = \frac{1}{1+x}$, $g(x) = \sqrt[3]{x}$, $h(x) = \frac{1}{x^3}$

37–42 Express f as a composition of two functions; that is, find g and h such that $f = g \circ h$. [Note: Each exercise has more than one solution.] ■

37. (a) $f(x) = \sqrt{x+2}$ (b) $f(x) = |x^2 - 3x + 5|$

38. (a) $f(x) = x^2 + 1$ (b) $f(x) = \frac{1}{x-3}$

39. (a) $f(x) = \sin^2 x$ (b) $f(x) = \frac{3}{5 + \cos x}$

40. (a) $f(x) = 3 \sin(x^2)$ (b) $f(x) = 3 \sin^2 x + 4 \sin x$

41. (a) $f(x) = (1 + \sin(x^2))^3$ (b) $f(x) = \sqrt{1 - \sqrt[3]{x}}$

42. (a) $f(x) = \frac{1}{1-x^2}$ (b) $f(x) = |5 + 2x|$

[Stewart]

2. Explain how the following graphs are obtained from the graph of $y = f(x)$.

(a) $y = 5f(x)$

(b) $y = f(x - 5)$

(c) $y = -f(x)$

(d) $y = -5f(x)$

(e) $y = f(5x)$

(f) $y = 5f(x) - 3$

3. The graph of $y = f(x)$ is given. Match each equation with its graph and give reasons for your choices.

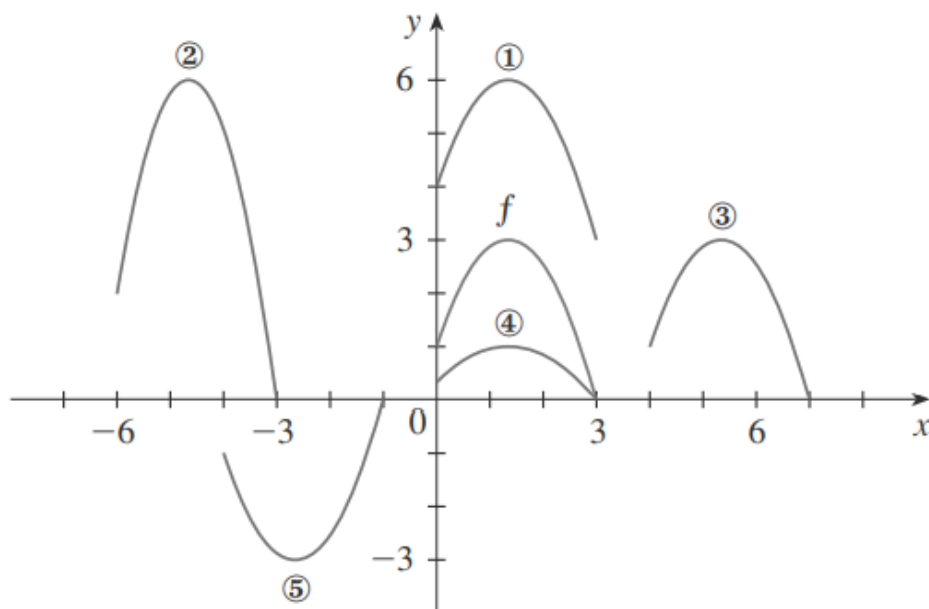
(a) $y = f(x - 4)$

(b) $y = f(x) + 3$

(c) $y = \frac{1}{3}f(x)$

(d) $y = -f(x + 4)$

(e) $y = 2f(x + 6)$



41–44 |||| Find $f \circ g \circ h$.

41. $f(x) = x + 1, \quad g(x) = 2x, \quad h(x) = x - 1$

42. $f(x) = 2x - 1, \quad g(x) = x^2, \quad h(x) = 1 - x$

43. $f(x) = \sqrt{x - 1}, \quad g(x) = x^2 + 2, \quad h(x) = x + 3$

44. $f(x) = \frac{2}{x + 1}, \quad g(x) = \cos x, \quad h(x) = \sqrt{x + 3}$

□ □ □ □ □ □ □ □ □ □

45–50 |||| Express the function in the form $f \circ g$.

45. $F(x) = (x^2 + 1)^{10}$

46. $F(x) = \sin(\sqrt{x})$

47. $G(x) = \frac{x^2}{x^2 + 4}$

48. $G(x) = \frac{1}{x + 3}$

49. $u(t) = \sqrt{\cos t}$

50. $u(t) = \frac{\tan t}{1 + \tan t}$

□ □ □ □ □ □ □ □ □ □

51–53 |||| Express the function in the form $f \circ g \circ h$.

51. $H(x) = 1 - 3^{x^2}$

52. $H(x) = \sqrt[3]{\sqrt{x} - 1}$

53. $H(x) = \sec^4(\sqrt{x})$

□ □ □ □ □ □ □ □ □ □

54. Use the table to evaluate each expression.

(a) $f(g(1))$

(b) $g(f(1))$

(c) $f(f(1))$

(d) $g(g(1))$

(e) $(g \circ f)(3)$

(f) $(f \circ g)(6)$

Exponential Functions:

In each of Exercises 7–10, sketch the shifted exponential curves.

- 7. $y = 2^x - 1$ and $y = 2^{-x} - 1$
- 8. $y = 3^x + 2$ and $y = 3^{-x} + 2$
- 9. $y = 1 - e^x$ and $y = 1 - e^{-x}$
- 10. $y = -1 - e^x$ and $y = -1 - e^{-x}$

[Stewart]

3–6 ||| Graph the given functions on a common screen. How are these graphs related?

- 3. $y = 2^x$, $y = e^x$, $y = 5^x$, $y = 20^x$
- 4. $y = e^x$, $y = e^{-x}$, $y = 8^x$, $y = 8^{-x}$
- 5. $y = 3^x$, $y = 10^x$, $y = \left(\frac{1}{3}\right)^x$, $y = \left(\frac{1}{10}\right)^x$
- 6. $y = 0.9^x$, $y = 0.6^x$, $y = 0.3^x$, $y = 0.1^x$

- 13. Starting with the graph of $y = e^x$, write the equation of the graph that results from
 - (a) shifting 2 units downward
 - (b) shifting 2 units to the right
 - (c) reflecting about the x -axis
 - (d) reflecting about the y -axis
 - (e) reflecting about the x -axis and then about the y -axis
- 14. Starting with the graph of $y = e^x$, find the equation of the graph that results from
 - (a) reflecting about the line $y = 4$
 - (b) reflecting about the line $x = 2$

Inverse functions:

In Exercises 7–10, determine from its graph if the function is one-to-one.

$$7. f(x) = \begin{cases} 3 - x, & x < 0 \\ 3, & x \geq 0 \end{cases}$$

$$8. f(x) = \begin{cases} 2x + 6, & x \leq -3 \\ x + 4, & x > -3 \end{cases}$$

$$9. f(x) = \begin{cases} 1 - \frac{x}{2}, & x \leq 0 \\ \frac{x}{x+2}, & x > 0 \end{cases}$$

$$10. f(x) = \begin{cases} 2 - x^2, & x \leq 1 \\ x^2, & x > 1 \end{cases}$$

Each of Exercises 25–36 gives a formula for a function $y = f(x)$. In each case, find $f^{-1}(x)$ and identify the domain and range of f^{-1} . As a check, show that $f(f^{-1}(x)) = f^{-1}(f(x)) = x$.

$$25. f(x) = x^5$$

$$26. f(x) = x^4, \quad x \geq 0$$

$$27. f(x) = x^3 + 1$$

$$28. f(x) = (1/2)x - 7/2$$

$$29. f(x) = 1/x^2, \quad x > 0$$

$$30. f(x) = 1/x^3, \quad x \neq 0$$

$$31. f(x) = \frac{x+3}{x-2}$$

$$32. f(x) = \frac{\sqrt{x}}{\sqrt{x}-3}$$

$$33. f(x) = x^2 - 2x, \quad x \leq 1$$

$$34. f(x) = (2x^3 + 1)^{1/5}$$

(Hint: Complete the square.)

[Howard Anton]

1. In (a)–(d), determine whether f and g are inverse functions.
 - (a) $f(x) = 4x$, $g(x) = \frac{1}{4}x$
 - (b) $f(x) = 3x + 1$, $g(x) = 3x - 1$
 - (c) $f(x) = \sqrt[3]{x-2}$, $g(x) = x^3 + 2$
 - (d) $f(x) = x^4$, $g(x) = \sqrt[4]{x}$
2. Check your answers to Exercise 1 with a graphing utility by determining whether the graphs of f and g are reflections of one another about the line $y = x$.
3. In each part, use the horizontal line test to determine whether the function f is one-to-one.
 - (a) $f(x) = 3x + 2$
 - (b) $f(x) = \sqrt{x-1}$
 - (c) $f(x) = |x|$
 - (d) $f(x) = x^3$
 - (e) $f(x) = x^2 - 2x + 2$
 - (f) $f(x) = \sin x$
4. In each part, generate the graph of the function f with a graphing utility, and determine whether f is one-to-one.
 - (a) $f(x) = x^3 - 3x + 2$
 - (b) $f(x) = x^3 - 3x^2 + 3x - 1$

9–16 Find a formula for $f^{-1}(x)$. ■

9. $f(x) = 7x - 6$
10. $f(x) = \frac{x+1}{x-1}$
11. $f(x) = 3x^3 - 5$
12. $f(x) = \sqrt[5]{4x+2}$
13. $f(x) = 3/x^2$, $x < 0$
14. $f(x) = 5/(x^2 + 1)$, $x \geq 0$
15. $f(x) = \begin{cases} 5/2 - x, & x < 2 \\ 1/x, & x \geq 2 \end{cases}$
16. $f(x) = \begin{cases} 2x, & x \leq 0 \\ x^2, & x > 0 \end{cases}$

17–20 Find a formula for $f^{-1}(x)$, and state the domain of the function f^{-1} . ■

17. $f(x) = (x+2)^4$, $x \geq 0$
18. $f(x) = \sqrt{x+3}$
19. $f(x) = -\sqrt{3-2x}$
20. $f(x) = x - 5x^2$, $x \geq 1$

[Stewart]

15–16 |||| Use a graph to decide whether f is one-to-one.

15. $f(x) = x^3 - x$

16. $f(x) = x^3 + x$

□ □ □ □ □ □ □ □ □ □ □

17. If f is a one-to-one function such that $f(2) = 9$, what is $f^{-1}(9)$?

18. Let $f(x) = 3 + x^2 + \tan(\pi x/2)$, where $-1 < x < 1$.

(a) Find $f^{-1}(3)$.

(b) Find $f(f^{-1}(5))$.

19. If $g(x) = 3 + x + e^x$, find $g^{-1}(4)$.

55–56 |||| Find (a) the domain of f and (b) f^{-1} and its domain.

55. $f(x) = \sqrt{3 - e^{2x}}$

56. $f(x) = \ln(2 + \ln x)$

Starting with the graph of $y = \ln x$, find the equation of the graph that results from

(a) shifting 3 units upward

(b) shifting 3 units to the left

(c) reflecting about the x -axis

(d) reflecting about the y -axis

(e) reflecting about the line $y = x$

(f) reflecting about the x -axis and then about the line $y = x$

(g) reflecting about the y -axis and then about the line $y = x$

(h) shifting 3 units to the left and then reflecting about the line $y = x$

23. If $f(x) = 2x + \ln x$, find $f^{-1}(2)$.

24. Find the inverse function of $f(x) = \frac{x+1}{2x+1}$.